COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-DBS-001: Supported Databases and Versions

Document ID	TS-DBS-001
Title	Supported Databases and Versions
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, including employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	4.01
Effective Date	07/01/2021
Last Updated	Not documented (Draft)

Purpose

OIT is statutorily mandated to provide IT support and services to the State and its stakeholders in the most effective and efficient manner possible (as defined by C.R.S. 24-37.5). To satisfy stakeholder needs surrounding OIT-supported databases and their versions, this document is submitted as the standard for specifications, methodologies, and practices to provide the service.

This document communicates the standards OIT has developed with the intent of providing enterprise standard configurations for supported databases and their currently supported versions, based on a best-of-value decision assessment involving factors such as stakeholder needs, hardware configurations serviced and supported by OIT, and operational and security best practices.

The ultimate purpose of this standard is to describe the minimum requirements for database support for the agencies, ensuring an easily supported environment that eases maintenance and minimizes style conflicts, while meeting minimum requirements for security and accessibility for government.

Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed “go-forward” or “net-new”, and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

Standards are predicated on the assumption that there can be no point of enforcement unless and until there is an effort to re-procure, renew (contracts), reconfigure (including upgrades), or redevelop the system. When an investment decision is made to sustain or improve the system, conformance to existing standards should be part of the assessed level of effort. For guidance or enforcement in conformance to standards, an email request may be sent to OIT_OEA@state.co.us with the subject line “Request for Architecture Review: <projectName/systemName>”.

For the purpose of enforcement, this standard is rated CAT 1 according to the Enforcement Category Chart.

Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

Additional Assignments

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

- Database Services Management: Responsible for enforcement and approval of changes to policies, standards, guidelines, and procedures.
- Database Services Database Administrator (DBA): Responsible for knowing and adhering to policies, standards, guidelines, and procedures as part of job description.

Standards

Goals

Ensure Database Services is able to provide support, patches, and upgrades specific to database platforms, versions, and environments.

Supported Architectures

- Specific supported architectures and versions are maintained in the IAW Technology Standards document.
- SQL Server - Current version minus 2 releases.
- Oracle - Current version minus 1 release.
- Informix - Maintain at the level referenced in the IAW Technology Standards document.
- AWS Cloud support - MySQL, Aurora DB, SQL Server, Oracle, DynamoDB, PostgreSQL, Redshift.
- Azure Power Platform Dataverse.

Requirements

Database Services will be involved in database version selection and database platform preferences.

Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the “Owner of Standard” listed under Roles and Responsibilities.

Exception requests may be acquired by completing the “EXCEPTION REQUEST to Technical Standard” form and including the completed form (with all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

References

- OIT documents and policies
- State of Colorado Style Library
- TS-OEA-001: Technology Accessibility for Persons with Disabilities
- Enforcement Category Chart
- IAW Technology Standards
- State documents and policies
- Colorado Brand Guidelines
- C.R.S. 24-37.5-101, et seq.

Definitions

- Production Database: State Agency database used for production purposes to conduct State business.
- Non-Production Database: State Agency database used for non-production purposes such as development, testing, staging, demonstration, or training.
- Business: Employees of supported State agencies, and their respective stakeholders, who use DBS-supported databases.
- SDS: Enterprise Application development group responsible for application development.
- OEA (Office of Enterprise Architecture): Formed as an office under OIT by Governor's Rule 8 CCR 1501-7.
- OIT (Office of Information Technology): Created and governed by C.R.S. 24-37.5, et seq.

- Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception, subject to review and approval by an authority in the area.